HK IMO Training 2019-2020
Problem Set 27
For 16 May 2020

Problem 105. Let $\triangle ABC$ be scalene with incentre I and with CN as the angle bisector of angle C. The line CN meets the circumcircle of $\triangle ABC$ again at M. The line ℓ is parallel to AB and touches the incircle of $\triangle ABC$. The point R on ℓ is such that $CI \perp IR$. The circumcircle of $\triangle MNR$ meets the line IR again at S.

Prove that $AS = BS$.

Problem 106. Find all functions $f\colon \mathbb{Z} \to \mathbb{Z}$ such that

$$f(4x+3y) = f(3x+y) + f(x+2y) \quad \text{for all } x, y \in \mathbb{Z}.$$

Problem 107. Find the maximum possible value of k for which in a 20-element set there are $2k+1$ different 7-element subsets such that each of these subsets intersects exactly k other subsets.

Problem 108. Let p be a prime number and a, k be positive integers such that $p^a < k < 2p^a$.

Prove that there exists a positive integer n such that

$$n < p^{2a} \quad \text{and} \quad \binom{n}{k} \equiv n \equiv k \pmod{p^a}.$$